The Aesthetics of Generation: Stylometric Analysis of LLM Identity

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Abstract

We investigate whether Large Language Models possess distinctive stylistic identities — recognisable "voices" that persist across tasks, genres, and explicit tone manipulation. Using a controlled corpus of 360 texts generated by GPT-4-Turbo, LLaMA 3, and Mistral under identical prompts and parameters across four genres, we extract 19 stylometric features spanning lexical diversity, readability, sentence structure, part-of-speech ratios, punctuation, and higher-level style. A Random Forest classifier with GroupKFold cross-validation (by prompt, to prevent leakage) achieves a macro-F1 of $\mathbf{0.751 \pm 0.042}$, more than double the three-class chance baseline. Paragraph-level features dominate model discrimination (**∼20%** of total RF feature importance). A leave-one-genre-out transfer experiment confirms that model fingerprints persist across genres (mean accuracy 0.558 vs. 0.333 chance). A dedicated robustness test on 144 tone-modified texts plus 48 neutral baselines reveals that casual tone is the most disruptive condition (accuracy 0.458, **−54.2** pp from neutral), while LLaMA 3 maintains the most persistent signature (accuracy 0.938–1.000 under casual and human-mimic) and Mistral the least robust (accuracy 0.062 under casual). Our results demonstrate that LLMs carry measurable aesthetic fingerprints and that multi-model attribution is feasible with interpretable stylometric features alone.

Keywords: stylometry, authorship attribution, large language models, text classification, discourse analysis

1 Introduction

Large Language Models (LLMs) have achieved remarkable fluency in text generation, raising a question that extends beyond performance benchmarks: do these models possess distinctive stylistic identities? Just as human authors develop recognisable voices through habitual choices of vocabulary, syntax, and rhetorical structure, the architectural design, training data, and decoding strategies of an LLM may leave characteristic imprints on every text it produces. We call these imprints *aesthetic fingerprints* and ask whether they are sufficiently consistent to permit *authorship attribution* — identifying which model generated a given text.

1.1 Stylometry and Authorship Attribution

Stylometry — the quantitative study of linguistic style — assumes that every author makes largely unconscious choices of vocabulary, syntax, and discourse structure that produce a measurable *stylistic fingerprint*. Modern systems extract features at multiple levels — lexical (type-token ratio, hapax rate), syntactic (sentence length, POS ratios, parse-tree depth), and discourse (connective usage, paragraph structure) — and feed them to machine-learning classifiers.

1.2 AI-Generated Text Detection

The rapid improvement of neural generators has spurred parallel work on detecting machine-generated content. Zellers et al. [1] showed that their Grover model could distinguish its own outputs from human writing with 92% accuracy and detect GPT-2 generations zero-shot with 96% accuracy, demonstrating that generation introduces statistical fingerprints at the distributional level.

Feature-based approaches offer complementary advantages: interpretability and cross-domain robustness. Opara [2] built StyloAI, extracting 31 stylometric features (including bigram/trigram uniqueness and Gunning Fog index) to train a Random Forest, achieving 81% accuracy on a 55,677-text corpus. In their setting, lexical-diversity measures were the most discriminative features, suggesting systematic differences in vocabulary balance between AI-generated and human-written text. Kumarage et al. [3] extended stylometric detection to short Twitter texts, where punctuation features were most discriminative and the signals appeared topic-invariant — a property that motivates our cross-genre transfer design.

1.3 From Binary Detection to Multi-Model Attribution

Much existing work frames the task as a binary decision: human versus machine. The scientific question is increasingly not *whether* a text was machine-generated, but *which* machine produced it — a shift that parallels the evolution of authorship attribution from forgery detection to candidate identification. Opara [2] framed the task as binary AI-vs-human detection, leaving multi-model attribution largely unexplored. Zellers et al. [1] showed that generated text carries detectable model-specific artefacts, suggesting that attribution among generators may be feasible. We address this gap directly, training a classifier to distinguish *among* GPT-4-Turbo [4], LLaMA 3 [5], and Mistral [6] using only interpretable stylometric features. Beyond standard cross-validation, we probe robustness with a cross-genre transfer test and a novel tone-manipulation experiment that asks whether explicit style instructions can defeat model attribution.

2 Research Question and Methodology

2.1 Research Questions and Problem Definition

We formally define the attribution task as follows. Let $\mathcal{M} = \{m_1, m_2, m_3\}$ denote the set of three LLMs (GPT-4-Turbo, LLaMA 3, Mistral) and let $\phi : \mathcal{T} \to \mathbb{R}^{19}$ be the stylometric feature map from a text $t \in \mathcal{T}$ to a 19-dimensional feature vector. The goal is to learn a classifier $f : \mathbb{R}^{19} \to \mathcal{M}$ that correctly attributes any text to its generating model. We pursue three research questions:

RQ1. Do different LLMs produce statistically distinguishable stylistic signatures?
RQ2. Which stylometric features best discriminate between models?
RQ3. How robust are these signatures across genres *and under explicit tone manipulation*?

2.2 Corpus Construction

To minimize topical confounds, we constructed a balanced, controlled corpus in which every model receives *identical prompts* under *identical generation parameters*.

Main corpus.

We designed 40 writing prompts (10 per genre) across four structurally distinct genres: **Narration** (short stories with sensory detail), **Argumentation** (essays with thesis and counter-arguments), **Dialogue** (conversations between characters), and **Description** (sensory-rich scene depictions). Topics were chosen to be neutral and diverse (travel, food, education, nature, etc.). Every prompt carried shared output constraints: 350–450 words, plain prose (no headings or markdown), and — critically — *no self-identification as an AI*, helping ensure that attribution relies on stylometric signal rather than trivial self-reference. Three contemporary LLMs were used: **GPT-4-Turbo** (OpenAI API), **LLaMA 3** (Meta, via Ollama), and **Mistral** (Mistral AI, via Ollama). All models used identical settings: `temperature` = 0.7, `top_p` = 1.0, `max_tokens` = 800. Each model generated three independent runs per prompt, yielding $3 \times 40 \times 3 = 360$ texts (120 per model, 90 per genre). All prompts — for both the main and robustness corpora — were initially drafted by GPT-5.3 (OpenAI) and subsequently revised and validated by the author to ensure genre-appropriateness, topic neutrality, and consistency of output constraints.

Robustness corpus.

To test whether explicit tone instructions can defeat model attribution, we constructed a second corpus using 8 base prompts (2 per genre) and three tone modifiers: **formal** (academic, structured), **casual** (relaxed, conversational), and **human-mimic** (instructed to write as a human would). Each model generated 2 runs per prompt–tone combination, yielding $8 \times 3 \times 3 \times 2 = 144$ tone-modified texts. We added 48 neutral baseline texts (the same 8 prompts, runs 1–2, unmodified) for a balanced design of 192 texts total (48 per condition).

Table 1: Stylometric feature set (19 features) used in all experiments.

Category	#	Features
Lexical diversity	6	type–token ratio, hapax legomena ratio, avg. word length, stop-word ratio, bigram uniqueness ratio, trigram uniqueness ratio
Readability	1	Gunning Fog index
Sentence / phraseology	4	avg. sentence length, sentence length std, paragraph count, avg. paragraph length
POS ratios	4	noun ratio, verb ratio, adjective ratio, adverb ratio
Punctuation	2	comma ratio, question mark ratio
Higher-level style	2	avg. syntactic depth, discourse marker total ratio

2.3 Stylometric Feature Extraction

We extracted 19 stylometric features using a custom pipeline built on spaCy (`en_core_web_sm`). Table 1 lists the full feature set across six categories, grounded in the feature inventories of Opara [2] and Kumarage et al. [3].

Lexical diversity features capture vocabulary richness: type–token ratio (TTR) and hapax legomena ratio measure diversity, while bigram and trigram uniqueness quantify local repetition. Stop-word ratio reflects grammatical habit rather than topical content. **Readability** is represented by the Gunning Fog index, which estimates cognitive complexity via syllable-counting heuristics. **Sentence-level** features operate at the phraseology level; paragraph count and average paragraph length capture document-level organisation. **POS ratios** — noun, verb, adjective, adverb — measure syntactic preferences normalised by token count. **Punctuation** counts (comma and question mark per sentence) capture formatting habits. **Higher-level style** includes average syntactic depth (mean token distance to the dependency root) and total discourse marker density (additive, contrastive, causal, temporal, and elaborative markers normalised by token count). All features were z-score normalised inside each cross-validation fold to prevent data leakage.

2.4 Classification and Evaluation

We trained a **Random Forest** classifier ($n_{\text{estimators}} = 1000$, all other parameters default), chosen for its native feature-importance scores, robustness to correlated features, and strong performance in prior stylometric work [2]. Evaluation follows a four-stage protocol:

1. **GroupKFold 5-fold cross-validation** by `prompt_id`: the same prompt never appears in both train and test, so the classifier must rely on model-specific style rather than prompt-specific content. Macro-averaged F1 is the primary metric.
2. **Group-aware 80/20 train–test split**: 288 training samples (32 prompts) and 72 test samples (8 prompts) with zero prompt overlap. Per-model precision, recall, and F1 are reported on this held-out set, and the trained model is saved for robustness evaluation.
3. **Cross-genre transfer** (leave-one-genre-out): for each genre, a fresh RF is trained on all texts from the other three genres and tested on the held-out genre. A mean transfer accuracy above the 0.333 chance baseline provides evidence that model fingerprints persist across genres.
4. **Tone-manipulation robustness**: the saved RF classifier predicts on the 192-text robustness corpus. Per-tone and per-model accuracy are compared against the neutral baseline.

3 Experimental Results

3.1 Exploratory Feature Analysis

Before classification, we examined how the 19 stylometric features distribute across models. Figure 1 shows violin plots for six representative features. Clear inter-model differences are visible in paragraph structure (`avg_paragraph_length`, `paragraph_count`) and lexical diversity (`type_token_ratio`), while other features such as `discourse_total_ratio` show more overlap. These distributional differences motivate a classification approach but also suggest that not all features will contribute equally.

To visualise the global structure of the 19-dimensional feature space, we applied PCA after z-score normalisation. Figure 2 shows PCA projections coloured by model and by genre. The model-coloured projection (a) reveals substantial overlap — the first two principal components do not cleanly separate the three models, suggesting that the discriminative signal is distributed across many dimensions

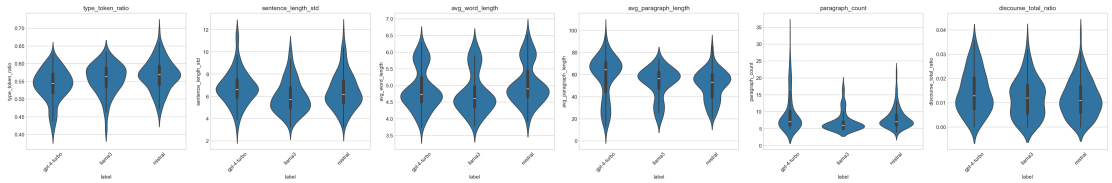
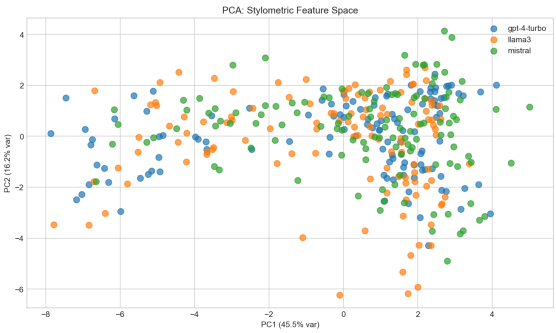
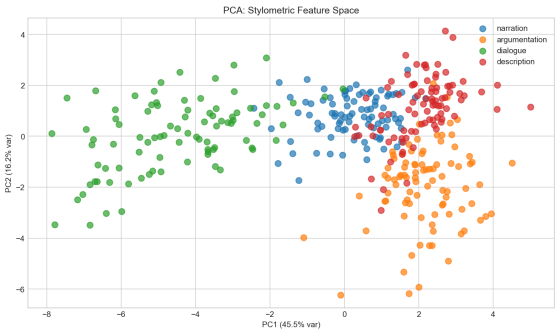


Fig. 1: Violin plots of six representative features across models. Paragraph-level features show the strongest inter-model separation.

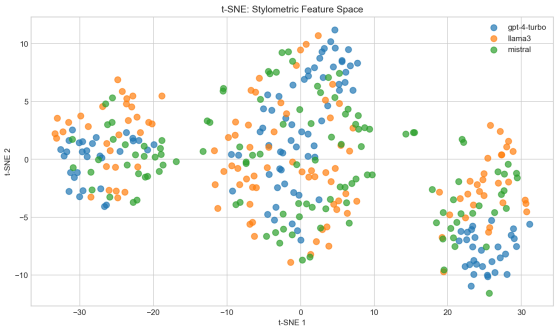


(a) By model: substantial overlap; linear projection does not cleanly separate models.

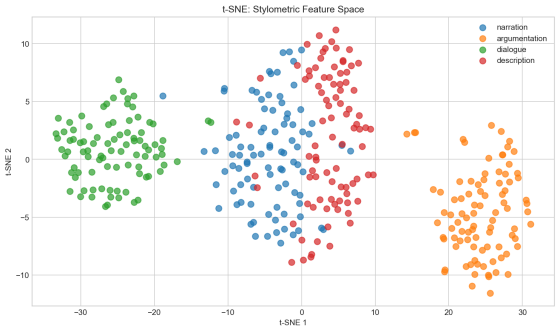


(b) By genre: genre-level clustering is visible, especially for dialogue.

Fig. 2: PCA projections of the 19-feature space (z-score normalised).



(a) By model: GPT-4-Turbo forms tighter clusters; LLaMA 3 and Mistral overlap.



(b) By genre: genre structure is clearly visible, especially for dialogue.

Fig. 3: t-SNE projections of the 19-feature space (z-score normalised).

rather than concentrated in a few. The genre-coloured projection (b) shows genre-related clustering (dialogue is particularly distinct), indicating that genre introduces variation that should be controlled for in classification.

Since PCA captures only linear structure, we also applied t-SNE to explore possible non-linear relationships in the stylometric feature space. Figure 3 shows t-SNE projections coloured by model and by genre. In the model-coloured projection (a), the three models remain substantially intermingled, although some local grouping is visible, suggesting that model-specific stylistic differences are present but appear weaker than genre effects. In contrast, the genre-coloured projection (b) shows clearer genre-related organisation, with relatively distinct regions for dialogue and argumentation and partial overlap between narration and description. This indicates that genre is a prominent source of variation, supporting the need to control for genre and motivating the cross-genre transfer test.

3.2 Classification Performance

GroupKFold 5-fold cross-validation yields a macro-F1 of 0.751 ± 0.042 (fold scores: 0.803, 0.777, 0.730, 0.680, 0.762), more than double the three-class chance baseline of 0.333 (**RQ1**).

Table 2 reports per-model results on the group-aware held-out test set ($n = 72$, 8 prompts, zero prompt overlap). Test accuracy is 0.764 and macro-F1 is 0.757. A clear performance asymmetry emerges: GPT-4-Turbo is the most distinctively voiced ($F1 = 0.85$), while Mistral is hardest to

Table 2: Per-model performance on the held-out test set (RF, 1000 trees, $n = 72$). Test accuracy: 0.764, macro-F1: 0.757.

Model	Precision	Recall	F1
GPT-4-Turbo	0.87	0.83	**0.85**
LLaMA 3	0.73	0.92	0.81
Mistral	0.68	0.54	0.60

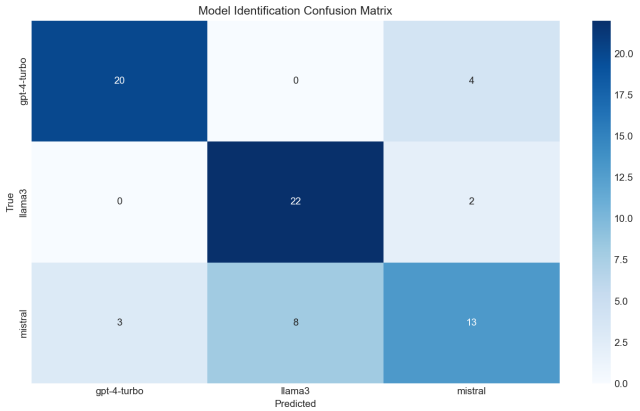


Fig. 4: Confusion matrix on the held-out test set. The primary confusion axis is between LLaMA 3 and Mistral.

Table 3: Top 10 features by RF feature importance (19 features, 1000 trees).

Rank	Feature	Importance	Category
1	`avg_paragraph_length`	0.115	Sentence / phraseology
2	`paragraph_count`	0.087	Sentence / phraseology
3	`trigram_uniqueness_ratio`	0.075	Lexical diversity
4	`avg_word_length`	0.066	Lexical diversity
5	`type_token_ratio`	0.060	Lexical diversity
6	`sentence_length_std`	0.059	Sentence / phraseology
7	`stop_word_ratio`	0.053	Lexical diversity
8	`pos_adv_ratio`	0.052	POS ratios
9	`pos_adj_ratio`	0.052	POS ratios
10	`avg_syntactic_depth`	0.051	Higher-level style

attribute ($F1 = 0.60$) and most frequently confused with LLaMA 3 (Figure 4). This is consistent with the t-SNE projection (Figure 3a), where LLaMA 3 and Mistral clusters overlap.

3.3 Feature Importance

Table 3 shows the top 10 features by RF feature importance. Two observations address **RQ2**.

First, **paragraph structure dominates**: `avg_paragraph_length` (0.115) and `paragraph_count` (0.087) are the two most important features, together accounting for ∼20% of total importance. The classifier relies most heavily on paragraph-related features, ahead of lexical and syntactic ones. Second, **lexical and syntactic features form a diverse second tier**: trigram uniqueness, average word length, type–token ratio, sentence length variability, and POS ratios jointly capture complementary dimensions of style. Importantly, all 19 features contribute meaningfully — no feature has near-zero importance (Figure 5), confirming the feature set contains no dead weight.

3.4 Cross-Genre Transfer

Table 4 shows leave-one-genre-out results (**RQ3**). All four genres exceed the chance baseline, providing evidence that LLMs carry a stylometric signal that persists across genre change. The mean transfer accuracy of 0.558 vs. 0.333 chance represents a substantial above-chance effect.

Narration transfers best (0.667), followed by description (0.622), argumentation (0.533), and dialogue (0.411). All four exceed the chance baseline.

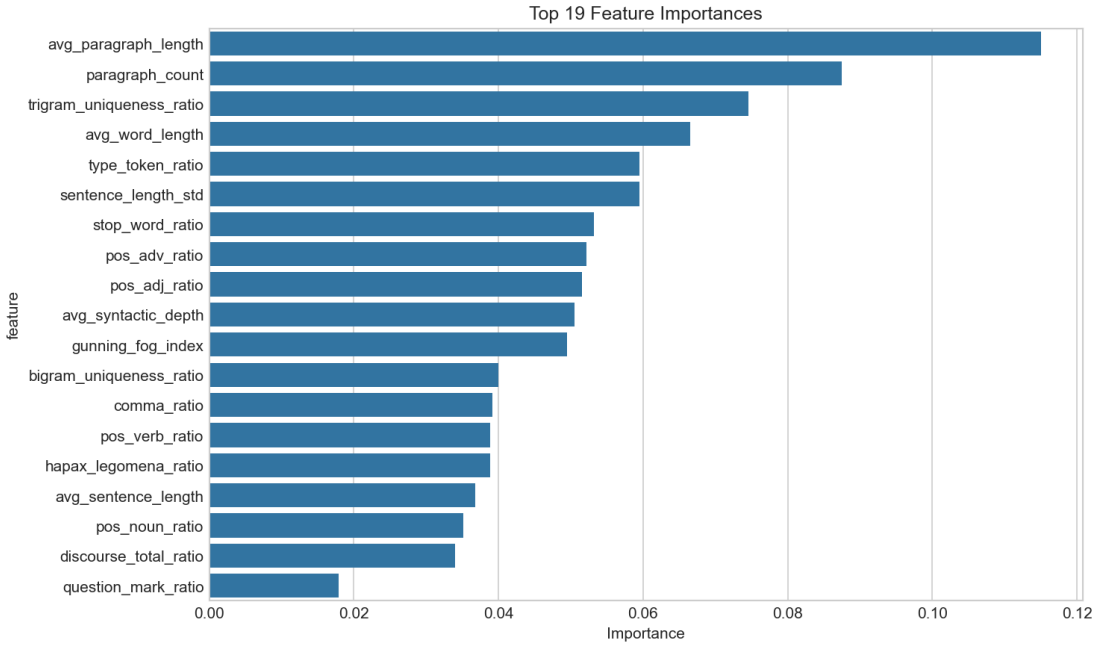


Fig. 5: All 19 features ranked by RF feature importance. Paragraph-level features dominate, followed by a diverse mix of lexical, syntactic, and punctuation features.

Table 4: Leave-one-genre-out transfer (train: 270 samples from 3 genres, test: 90 samples). Chance: 0.333.

Held-out genre	Accuracy	Macro-F1
Narration	**0.667**	0.663
Description	0.622	0.636
Argumentation	0.533	0.440
Dialogue	0.411	0.362
Mean	**0.558**	**0.525**
Chance baseline	0.333	—

Table 5: Robustness test: classification accuracy by tone condition and by model. Neutral baseline is in-sample (training prompts). Chance: 0.333.

	Neutral	Formal	Casual	Human-mimic
Overall accuracy	1.000	0.646	0.458	0.604
Drop from neutral	—	−35.4 pp	−54.2 pp	−39.6 pp
GPT-4-Turbo	1.000	0.750	0.375	0.500
LLaMA 3	1.000	0.750	**0.938**	**1.000**
Mistral	1.000	0.438	0.062	0.312

3.5 Robustness Under Tone Manipulation

The saved RF classifier (trained on the main corpus) is applied to the 192-text robustness corpus. Table 5 reports per-tone overall accuracy and per-model accuracy breakdowns.

All tone conditions remain above the three-class chance baseline (0.333), but accuracy drops substantially from the perfect neutral baseline (1.000, in-sample). **Casual tone is the most disruptive** condition, reducing accuracy by 54.2 percentage points. Formal tone is the least disruptive (−35.4 pp), suggesting that structured, academic prose may preserve more of each model's learned patterns than relaxed, conversational writing. Human-mimic falls between the two (−39.6 pp), closer to formal than to casual.

At the model level, **LLaMA 3 shows the most persistent stylistic fingerprint**: even under explicit tone manipulation, the classifier identifies it with 93.8–100% accuracy (casual and human-mimic). Under formal tone it matches GPT-4-Turbo at 75%. Its signature appears comparatively resistant to the tone instructions tested here. **Mistral is the least robust model**: under casual tone

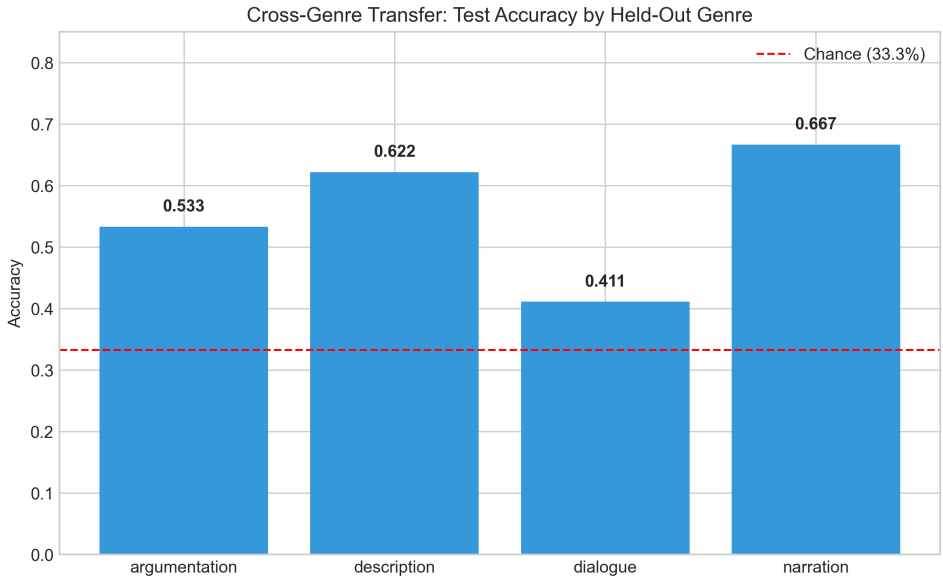


Fig. 6: Cross-genre transfer accuracy by held-out genre. Dashed red line marks the chance baseline (0.333). All genres exceed chance; narration transfers best, dialogue worst.

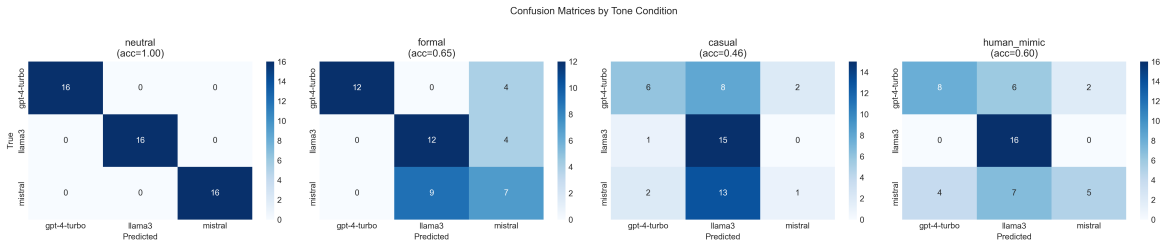


Fig. 7: Confusion matrices by tone condition ($n = 48$ each). Under casual tone, both Mistral and GPT-4-Turbo are predominantly misclassified as LLaMA 3, while LLaMA 3 itself remains correctly identified.

its accuracy falls to 6.2% — well below chance — indicating that casual tone strongly disrupts the classifier's ability to recover Mistral's baseline signature. The confusion matrices (Figure 7) reveal that under casual prompting, both Mistral and GPT-4-Turbo are predominantly misclassified *as LLaMA 3*, while LLaMA 3 itself remains correctly identified. One interpretation is that casual prompting moves these models into the feature region the classifier has learned to associate with LLaMA 3.

The mean accuracy across tone-modified conditions (0.569) is close to the cross-genre transfer accuracy (0.558). Both are distribution-shift scenarios, and in this project both yield performance in roughly the 0.55–0.60 range for the 19-feature RF classifier.

4 Concluding Remarks

4.1 Discussion

Our results provide affirmative answers to all three research questions.

RQ1: The RF classifier achieves macro-F1 of 0.751 ± 0.042 in GroupKFold cross-validation and 0.757 on the held-out test set, more than double the chance baseline. LLMs produce measurably distinct stylistic distributions. **RQ2**: Paragraph-level features — especially `avg_paragraph_length` and `paragraph_count` — are the dominant discriminative signal (∼20% of RF feature importance), and all 19 features contribute meaningfully. **RQ3**: Model fingerprints persist across genres (mean transfer accuracy 0.558 vs. 0.333 chance) and survive explicit tone manipulation (mean tone-modified accuracy 0.569), though casual tone is the most effective at disrupting attribution.

The performance asymmetry across models invites interpretation. GPT-4-Turbo ($F1 = 0.85$) is more attributable than LLaMA 3 (0.81) or Mistral (0.60), indicating that the three models do not express equally distinctive stylometric signatures under our experimental conditions. LLaMA 3 and Mistral are more frequently confused with each other than with GPT-4-Turbo, which may reflect closer stylistic proximity between the two open-weight models.

The dominance of paragraph-level features corroborates this interpretation: GPT-4-Turbo tends to produce more numerous and longer paragraphs, while LLaMA 3 uses fewer, more compact blocks. These are the kinds of structural choices that traditional stylometry often treats as authorial signals. That they persist under explicit tone manipulation — particularly for LLaMA 3 — suggests that paragraph-structure habits are resistant to surface-level prompting, though further work would be needed to determine whether this reflects architectural constraints or training-data patterns.

The convergence of the cross-genre transfer band ($\sim$0.558) and the tone-manipulation band ($\sim$0.569) is noteworthy. Both represent scenarios in which the input distribution shifts away from the classifier's training conditions, yet performance remains roughly 1.7$\times$ chance. In this project, these two distribution-shift settings produce similar above-chance accuracy levels.

Mistral's sharp accuracy drop under casual tone (6.2%) contrasts with LLaMA 3's resilience (93.8%). We observe the collapse but cannot isolate its cause from our data alone. Several non-exclusive hypotheses merit future investigation: (i) Mistral may follow tone instructions more faithfully, abandoning its default patterns to a greater extent; (ii) Mistral's baseline signature was already the weakest ($F1 = 0.60$), making it easier to override; (iii) when pushed into casual mode, Mistral's output shifts into LLaMA 3's feature region, possibly because casual tone removes the stylistic differences that distinguish them under neutral conditions.

4.2 Limitations

Several limitations bound the scope of these conclusions. First, our corpus of 360 texts (main) and 192 texts (robustness) is substantially smaller than StyloAI's 55,677-text dataset [2]; the controlled design gains experimental validity (identical prompts and parameters) at the cost of statistical power. Per-model robustness estimates (16 texts per cell) have wide confidence intervals. Second, we evaluated only three models; extending to Claude, Gemini, or smaller open-source variants would test scalability. Third, generation parameters were fixed at a single setting (`temperature` = 0.7, `max_tokens` = 800, `top_p` = 1.0); varying these would help disentangle architecture-driven style from sampling-strategy artefacts. Fourth, the neutral baseline in the robustness test uses training prompts, so 100% accuracy is expected and serves only as a reference point for tone comparisons, not as a held-out performance estimate. Finally, tone manipulation was implemented via user-prompt instructions; fine-tuned style transfer would be a stronger test of signature persistence.

4.3 Future Work

Several extensions are promising. Larger, more diverse corpora would test generalisability, while varying temperature and top-p within a single model could isolate decoding-strategy contributions to stylistic fingerprints. Neural-stylometric hybrids — concatenating our 19 features with transformer embeddings as in Kumarage et al. [3] — may improve accuracy while retaining partial interpretability. A temporal dimension — tracking how a model's fingerprint evolves across successive releases (e.g., GPT-3.5 $\rightarrow$ GPT-4 $\rightarrow$ GPT-4-Turbo) — could illuminate how training and alignment procedures shape machine "voice" over time.

Declarations.

AI Usage Disclaimer.

Portions of this manuscript were drafted and revised with the assistance of AI-based tools (ChatGPT and Claude). The same tools were used to assist with code development, revision, and debugging. Writing prompts for both corpora were initially drafted by GPT-5.3 and subsequently revised and validated by the author. All AI-generated content — text, code, and prompts — was critically reviewed, verified, and edited by the author. The experimental design, data analysis, and scientific conclusions are the sole responsibility of the author.

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